

Topic Modeling using Language Models

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TABLE OF CONTENTS

1. Plan.....	i
2. Introduction.....	ii
2.1. Structure	ii
3. Problem Definition.....	iii
4. Solutions.....	iv
5. Conclusion.....	ix
5.1. Challenges.....	ix
5.2. Open problems and future work	ix
REFERENCES	x

1. Plan

Research plan consists of six parts. In this plan, I aimed to get sufficient information about research topic and ultimately apply this information in hands on practice. Below you may find the task plan. Code practice task constitutes most of this research. Essentially, this research is based on comparison of legacy models, embedding based models and prompt engineering approaches.

Task	Deadline
Surveying Classical Topic Modeling	
Submit Progress Report	19.04.2026
Survey LLM-based topic modeling	26.04.2026
Survey Tools and Applications	03.05.2026
Code Practice	14.05.2026
Submit Final Report	17.05.2026

2. Introduction

Topic modeling is used to extract meaningful topics from text and group documents accordingly¹. Especially, large documents are particularly hard to group, which may be significant problem in real-world cases. Topic modeling addressess this by discovering the hidden patterns in text without manual grouping. Moreover, it handles this task through several different approaches. Classical methods such as Latent Dirichlet Allocation(LDA)[1] have been widely used for this purpose. In this research, I compare three different approaches in topic modeling. With the rising popularity Large Language Models, topic modeling – like every other modeling field - is also evolving, with newer approaches such as BERTopic [2] and TopicGPT [3] emerging. However, following popularity alone may be mistake, so comparing well-established techniques with LLM-based approaches is what motivated me to conduct this research.

2.1. Structure

This research report consists of five different sections. Section 1 explains the plan that I followed through my research. Section 2 provides an introduction to the topic and the motivation behind the research. Section 3 defines the problem by discussing the limitations of classical topic modeling methods and need for new approaches. Section 4 presents solution through examples from code practice that I compared three different approaches to topic modeling. Section 5 concludes the report with the main finding, challenges and future works. Overall structure of the report is summarized in Figure 2.1

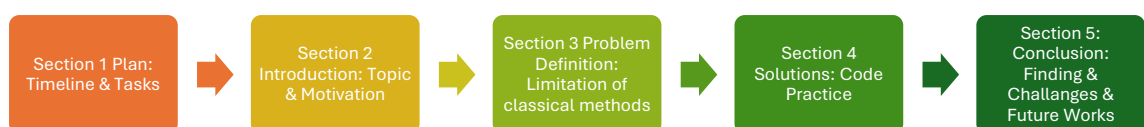


Figure 2.1: Structure of report

3. Problem Definition

Main aim of topic modeling is discovering themes in large text collections automatically. This research compares three different approaches to solve this problem:

Latent Dirichlet Allocation(LDA) is a model that assumes every document is a mixture of a small number of hidden topics. Also it treats each topic as a distribution over words. This method is based on bag-of-words representation which is a way of representing text as a number so that algorithms can work with it and require number of topics to be specified in advance.

BERTopic is a modular topic modeling framework that converts documents into semantic embeddings. After clustering these embeddings, it extracts topic label using TF-IDF which is a method that calculates importance of a word in a document. Unlike LDA, it captures meaning of a word and context rather than only counting frequency.

Prompt-based approaches use Large Language Models to generate readable topic labels for documents. They include different prompt engineering techniques such as Persona, CoT and few-shot which improve the performance of LLMs when they are combined carefully. In this research, a combination of these prompt engineering techniques was implemented using Gemini 2.5 Flash Lite model, accessed through PydanticAI agentic development framework.

The problem is that despite availability of these approaches, their comparison in practice remains unclear. LDA and BERTopic both produce word-list outputs, but they differ from their techniques on word selection. However, prompt-based LLMs produce readable natural language labels instead of word lists. Each approach introduces different trade-offs in interpretability, time and cost. The main goal of this research is to compare these three different approaches under same conditions.

4. Solutions

4.1. Code Practice Setup

This section describes the setup which I have used during comparison the three topic modeling approaches under the same condition.

These practices use the 20 Newsgroup dataset which is generally used benchmark for text classification and topic modeling. It includes 18,846 documents organized into 20 categories covering religion, sports, computer hardware, politics and science. The dataset is accessed through the scikit-learn library¹.

Compared Methods

Approach	Method	Library / Model
Classical	LDA	scikit-learn
Embedding-based	BERTopic	BERTopic (sentence-transformers)
Prompt-based	LLM prompting	Gemini 2.5 Flash-Lite + PydanticAI

¹ https://scikit-learn.org/stable/datasets/real_world.html#newsgroups-dataset

For LDA and BERTopic, dataset is fully used which consist of 18.846 documents. For prompt based approach, a sample of 200 documents(10 documents per category) was picked due to API rate limits and processing cost. This sampling ensures that every category is evaluated and represented in presentation equally.

The code is seperated into different modules for maintainbality and being able to see outputs clearly. A shared dataset module downloads the data and stores to prevent downloading again by different models. Each script produces both human-readable text file and machine-readbale JSON files.



Figure 4.1 Overall Code Practice Setup

4.2. Implementation

Latent Dirichlet Allocation was implemented using scikit-learn. Documents were converted into document-term matrix by CountVectorizer with English stop-word removal by passing as parameter. The LDA model was fitted with 10 topics. The top 10 words from each topic were extracted as topic representation.

BERTopic approach didn't need much code because it handles embedding, clustering and topic extraction internally so these steps don't need to be configured manually. The model was fitted on full dataset and produced comparable number of topics. Moreover, BERTopic generates an interactive visualization of the resulting topics.

Prompt-based approach implemented by using Gemini 2.5 Flash-Lite model and PydanticAI framework. A structured output model was defined so that the model returns topic label, a confidence score and three keywords for each document. System prompt combined three different prompt engineering techniques: a persona, Chain of Thought reasoning, few-shot examples. Some of the API calls returned errors due to API rate limits so a small number of the documents couldn't be sent to LLMs for topic extraction.

```
# output schema
class TopicResult(BaseModel):
    topic: str = Field(description="Short topic label (2-4 words)")
    confidence: float = Field(description="Confidence score between 0.0 and 1.0")
    keywords: list[str] = Field(description="Three key words from the text")
```

4.3. Results and Comparision

This part presents the result of three different approaches and compares them in terms of output format, topic quality, processing time and scalability.

Performance Comparision

Metric	LDA	BERTopic	Prompt-based
Documents processed	18.846	18.846	200(sample)
Number of topics	10	9	per-document
Processing time	48.6s	263.2s	476.9s
Output format	Word list	Word list	Natural language label
Preprocessing	Manual (stop words)	Integrated	Not required
Monetary cost	Free	Free	It has costs

Table 4.3: Performance comparison of the three approaches

Topic Quality

To be able to evaluate three approaches fairly, same documents were examined across all methods using the topic which each document was assigned to. Following table shows representative examples.

True Category	LDA Topic	BERTopic Topic	Prompt-based Label
soc.religion.christian	people,don,does,think,just	(outlier, no topic)	"Christian Discussion Group"
sci.electronics	space,research,nasa,university	radar,ground,detector,radio	"Amateur Radio Literature"
rec.autos	file,edu,image,com,program	people,dont,god,think	"Automotive Mailing List Setup"
rec.sport.baseball	just,don,like,good,think	game,team,players,season	"Baseball Player Analysis"
sci.med	just, don, like, good, think	<i>(outlier, no topic)</i>	"Allergy Shot Effectiveness"
talk.politics.guns	god, people, jesus, said	people dont,god,think	"Gun Control Legislation"

Table: How the same documents were classified by each approach.

LDA's output were unclear. Some were clearly interpretable in word lists but such as a religion document (soc.religion.christian) was assigned a unrelated topic dominated by "people,don,think,just". An automotive document (rec.autos) was placed in a topic about "file,edu,image,program" and a baseball document was assigned to the generic "just, don, like, good" topic. This shows that LDA's performance in document-to-topic assignments are often inaccurate.

BERTopic produced more coherent and thematically focused topics than LDA. However, it frequently used outlier category. When BERTopic assigned a topic, it was usually relevant. For example, the electronics document was correctly placed in a "radar, detector, radio" topic and the baseball document in a "game, team, players, season" topic. Although it marked several documents as outliers, it increases the precision because it doesn't force every document into a topic.

Prompt based approach produced accurate and readable labels for each processed document. Each document match the category and label with no generic or noise assignment and no unclassified documents.

Discussion

The results reveal clear trade-off between interpretability and cost. LDA is fastest but it fails in document-to-topic assignment and it requires manual interpretation. BERTopic offers significantly better topic coherence in five times longer waiting time and it still requires word-list interpretation though. The prompt-based approach produces the most directly usable output but it is the slowest among these approaches. It has costs and couldn't be scaled to full dataset due to API constraints.

No single method is universally superior. The choice must be done according to use-case: LDA for speed and large-scale exploration, BERTopic for coherent topics without manual interpretation cost and prompt-based methods when human readable labels are the priority and dataset is suitable and small.

5. Conclusion

This research compared three different approaches to topic modeling. LDA, BERTopic and Prompt-based approaches were compared on the 20 Newsgroups dataset. The result shows that there is a clear trade-off between speed, accuracy and interpretability. LDA was fastest but produced inaccurate output in document-to-topic assignment. BERTopic created better output than LDA but left some documents as unclassified outliers. The prompt-based approach produced the most accurate output. However it was the slowest and may have significant costs with huge datasets. Also, prompt-based may face some API constraints with some LLMs. Overall, no single method proved universally superior. The choice depends on specific use cases.

5.1. Challenges

BERTopic is sensitive to preprocessing because stop-word filtering significantly affected topic quality. API rate limits prevented the prompt-based method from processing full dataset.

5.2. Open problems and future work

The prompt-based approach could be tested with other LLMs to compare how different models perform on the same task. Also, additional classical methods such as LSA and NMF could be included into research and outputs can be evaluated in broader environment. As a last, prompt-based method can be applied to full dataset rather than a sample, it will allow cost analysis of LLM based topic modeling.

REFERENCES

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